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விவசாயத் திணைக்களம்
Department of Agriculture

Tomato - Disease Management

- **Fungal diseases**

1. Damping-off

Causal
Pathogens: *Phythium* spp.,*Phytophthora* spp.,*Fusarium* spp.,*Rhizoctonia* spp.,*Sclerotium* s pp., *Colletotrichum* spp Symptoms: Seedling death due to root infection

Disease Management:

- Virgin lands for nursery beds preparation
- Heat sterilization
- Seed Treatments – Thiram 80wp 5g/1Kg or Homai 4g/1kg
- Soil treatments – Thiram 80wp 7g/5 l/m^{-2} or Homai 5g/5 l/m^{-2}

2. Early blight

Causal Pathogen: *Alternaria solani* **Symptoms**: Dark Patches with concentric rings on leaves, fruits and stems

Management:

- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)

Mancozeb 80% WP – 20g/10l Metelaxyl 8%+ Mancozeb 64% WP -12.5g/10l Metiram 55% + Pyraclostrobin 5% WG – 8ml/10l

3. Late blight

Causal Pathogen: *Phytophthora infestans* **Symptoms**:

- First appear on the lower, older leaves as water-soaked grey-green spots.
- Later spots darken and a white fungal growth form on underside

Management:

- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)

Mancozeb 80% WP – 20g/10l Metelaxyl 8%+ Mancozeb 64% WP -12.5g/10l Metiram 55% + Pyraclostrobin 5% WG – 8ml/10l

4. Target Spot of Tomato caused by *Corynespora cassiicola*

Causal Pathogen: ***Corynespora cassiicola*** **Symptoms**

- Circular, dark brown necrotic lesions in a concentric pattern on leaves and surrounded by a yellow halo and spread to all the leaflets
- Leaves turn yellow and rapidly collapse and die
- Spots also occur on the stem and fruits

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Management: Suitable fungicides under testing

5. Powdery mildew

Causal organism: *Oidium lycopersicum* **Symptoms:**

- Light green to bright yellow lesions on the upper surface of the leaf
- Lower surface develop white powdery patches when the fungus sporulation
- Severe infection results in leaf defoliation

Management:

- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)
    - Apply sulfur 80% WG 50 g/10 l of water
    - Chlorothalonil 500g/l SC- 30 ml/10 l of water
    - Carbendazim 50% WP- 7 g/10 l of water

6. Anthracnose

Causal organism: *Colletotrichum coccodes* **Symptoms**

- Die-back from apical buds, Flower dropping
- Black , sunken leaf, stem and fruit lesions
- Fruit rot

Disease Managements:

- Seed treatment with recommended fungicides before seed sowing
- Use recommended fungicides at initial stage of disease development (Refer: Pesticide recommendation DOA, 2019)

7. Septoria leaf spot

Causal organism: *Septoria lycopersici*

Symptoms: Water-soaked areas on affected leaves become circular with brown to gray centers.

Control: Use recommended fungicide

Daconil	15- 30 ml/10L
Mancozeb	20 ml/10L
Topsin	6g/10L
Carbandazim	7g/10L

8. Collar rot/Root rot

Causal organism: *Fusarium* spp., *Rhizoctonia* spp., *Sclerotium* spp.

Disease symptoms:
The pathogen attacks the collar portion of plant; The infection leads to the drying of lower

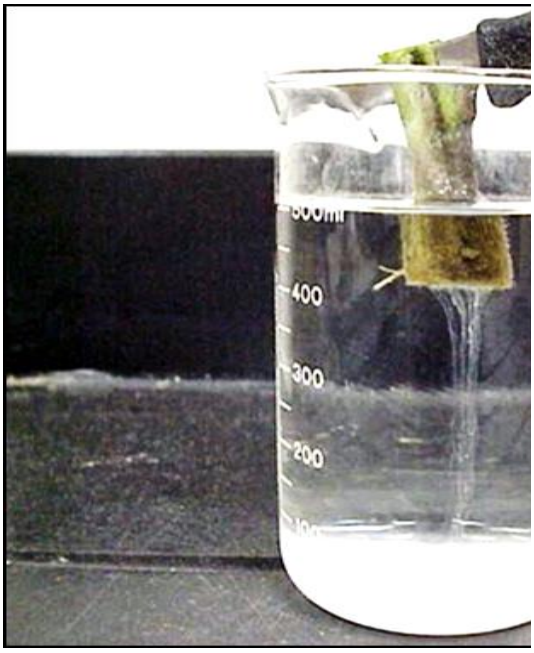
leaves and eventually the whole plant dries giving a typical symptom of wilting which ultimately leads to its death.

- **<u>Bacterial diseases</u>**

9. Bacterial wilt

Causal organism: *Ralstonia solanacearum*
Disease symptoms

- Permanent wilting of the plant during optimum water level in the soil
- Viscous ooze exudes at the cut end when immersed in water



Management

- Use wilt resistant varieties
- Crop rotation with legumes, cereals etc.
- Maintain Field sanitization

- **<u>Viral diseases</u>**

10. Tomato Yellow Leaf Curl Virus (TYLCV)

TYLCV transmitted by whiteflies.

Symptoms

- Upward curling of leaves
- Yellow (chlorotic) leaf margins
- Smaller leaves than normal
- Plant stunting and flower drop
- If tomato plants are infected early in their growth, there may be no fruit formed
- Infected plants may appear randomly throughout the Field

Prevention & Treatment:

- Eliminating weeds
- Remove the old crop debris

Virus disease Management

- Use disease free seeds
- Weed free field and field sanitation

- Reduce vector population (By insecticides)

11. Curly Top Virus (CTV)

Transmitted by : Leaf hoppers

- Inward rolling of leaflets along the midrib
- Petiole and midrib frequently curve downwards, giving the leaf a drooping but not wilting appearance
- Leaves are thick, crisp, brittle and dull green in color with purple veins

Prevention & Treatment:

- Eliminating weeds
- Remove the old crop debris

Virus disease Management

- Use disease free seeds
- Weed free field and field sanitation
- Reduce vector population (By insecticides)

12. Tomato Spotted Wilt Virus (TSWV)

TSWV is spread by a tiny insect called thrips

Symptoms

- Plants show bronzing, curling, necrotic streaks and spots on the leaves
- Dark-brown streaks also appear on leaf petioles, stems and growing tips
- Plants are small and stunted
- The ripe fruit shows paler red or yellow areas on the skin
- Plants are killed by severe necrosis

Prevention & Treatment:

- Eliminating weeds
- Remove the old crop debris

Virus disease management

- Use disease free seeds
- Weed free field and field sanitation

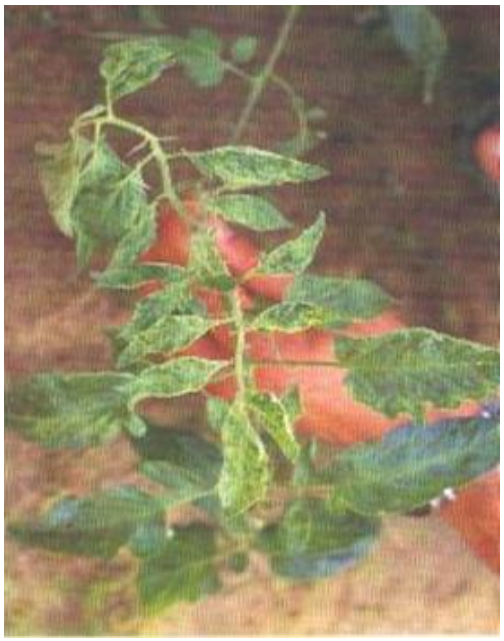
- Reduce vector population (By insecticides

13. Cucumber Mosaic Virus (CMV)

CMV transmits by aphids

Symptoms:

- Early stages are yellow, bushy and considerably stunted. Later, filiformity or shoestring-like leaf blades
- Shoestring like leaf blades
- Leaf mottling



Virus disease management

- Use disease free seeds
- Weed free field and field sanitation
- Reduce vector population (By insecticides)